

Evaluating the Economic and Environmental Impacts of California’s Cap-and-Invest Initiative

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Abstract

California’s cap-and-invest program, spearheaded by the California Air Resources Board (CARB), is a market-oriented climate change policy that seeks to reduce greenhouse gas emissions while limiting negative effects on economic growth. This paper evaluates if the program was successful in reducing carbon emissions, and if it affected growth along the way. We employ a synthetic control method, complemented by a differences-in-differences approach and placebo-based inference to execute this analysis. We find limited evidence that carbon emissions were significantly reduced by the program, and no statistically significant impact on real GDP growth.

Introduction

California’s cap-and-invest program (formerly cap-and-trade) is one of the core strategies in part of a larger effort by the California Air Resources Board and state government to reduce emissions across the state. Its implementation, alongside several other policies signed into law in the late 2000s and early 2010s, come as climate change and global warming become defining issues for many Americans. Many national and subnational governments turn to market-based approaches like cap-and-invest to reduce emissions while minimizing adverse effects in the economy. The success of these market-based approaches remains hotly contested among activists and academics, with many arguing these programs are not robust enough to deliver sufficient reductions in warming. This suggests a strong need for additional research into the subject.

Evaluating a causal effect of a single-state policy like this proves difficult due to the lack of a reliable counterfactual and the existence of policies with similar goals being implemented around the same time. This paper intends to contribute to this ongoing line of research and address difficulties in measurement by using a hybrid synthetic control model and differences-in-differences approach (the former proposed by Abadie and Gardeazabal (2003)) to build a reliable counterfactual in evaluating the existence of a causal effect on carbon emissions from cap-and-invest. The synthetic control model is particularly well-equipped for this task, as it approximates Cali-

fornia’s pre-treatment trajectory using a pool of donor states. We ask the question, did the cap-and-invest program implemented in 2013 cause a significant reduction in carbon emissions, and if it came at the cost of economic growth. The following experiment finds that there is little evidence of a substantial causal effect on emissions, though pre-treatment divergence suggests one should exercise caution in interpreting causal estimates. We combine this with an analysis of the program’s effect on economic growth, where we conclude similarly that there is insignificant evidence that growth was at all harmed or changed by climate policy.

The rest of the paper is structured as follows: Section 2 will cover the existing literature on the subject, Section 3 outlines the data collected, Section 4 the methodology, Section 5 presents the results, and Section 6 concludes the paper.

Literature Review

Literature on carbon pricing in general is quite exhaustive, with a wide array of work evaluating programs on the European Union’s Emissions Trading System and carbon markets in the United States and Canada. They seek to answer the following: whether carbon pricing effectively reduces emissions, and if it comes with a pronounced economic cost. Existing papers find modestly significant emissions reductions due to carbon pricing, but the results vary wildly depending on policy design, region, and sectoral coverage. Papers

on California’s cap-and-invest specifically is relatively limited.

Christian Lessmann and Niklas Kramer (2024) employ a synthetic counterfactual to evaluate the effect of cap-and-trade and sectoral emissions. They find that emissions in the power sector fall well below counterfactual emissions by 48 percent, while industrial sector emissions were actually 6 percent higher. They conclude that the overall slate of climate policy in California has largely failed to deliver effective decarbonization in the industrial sector. Emission reductions in the power sector are attributed to a switch from natural gas to renewable energy across the state. This approach paints a clear picture of where California’s climate legislation succeeded, and where it failed to deliver.

While the former focuses on a sectoral level using a synthetic control approach, Danilo Mascia and Enrico Onali (2023) choose to evaluate regional emissions instead, while also accounting for any reductions in economic growth. They come to the conclusion, using a differences-in-differences analysis, that there is little evidence of a reduction in either regional emissions or economic growth due to cap-and-invest. They emphasize the importance of designing climate policies at a regional level and subsequently evaluating their effects. they also find that interstate organizations do not transfer their emissions to facilities located in California, but parent-level concentration in emissions is changed.

The following paper aims to contribute to this existing literature by utilizing a synthetic control and differences-in-differences hybrid model to evaluate

cap-and-invest's effect statewide emissions and economic growth. The synthetic control is of particular use to us as using another state or group of states as a control is difficult; there are few states that closely match California's demographic and economic composition. This approach aims to provide a comprehensive assessment of the cap-and-invest program on a state level.

Data

The Bureau of Economic Analysis (BEA) is a U.S. government agency that measures economic information such as Gross Domestic Product (GDP) and personal income. Tables on said information can be filtered by year and region. Figures of interest to me are *SAGDP1* (state annual gross domestic product summary), where I obtained real GDP growth and totals per year, and *SAINC1* (personal income) which includes population estimates for each state. I calculated GDP per capita by dividing each real state GDP by the state population. This monetary data is all measured in real dollars chained to 2017. Emissions are measured in MtCO₂e, or CO₂ equivalent. GDP growth is measured in percentage change. It is important to note that GDP per capita is used in the predictor matrix, while GDP growth is an outcome variable. *SAGDP2* from the same set of tables was used to obtain the percentage of GDP that was generated by the manufacturing sector in each state. The manufacturing share, denoted by *mfg_share*, will be a predictor for our synthetic model.

The energy mix was obtained from the Energy Information Administration (EIA), a federal agency responsible for collecting data on energy sources. The energy mix represents the share of total electricity generated from each source in a given state. It is divided into subcategories *coal_share*, *gas_share*, *petrol_share*, and *renewable_share* and will be used in the predictor matrix that constructs the synthetic control model. This way, differences in energy production in each state are accounted for when building the model. Total carbon emissions by state was also taken from the EIA. *Emissions* will function as our primary outcome variable of interest. All data spans 2000-2023, with the treatment year defined as 2013, when cap-and-invest went into effect.

Methodology

To evaluate the costs and benefits of California’s cap-and-invest regime, I employ a synthetic control and a differences-in-differences hybrid to model the pre-treatment trajectory of California’s carbon emissions and predict what direction they would go without policy implementation. The treatment in this case is the implementation of cap-and-invest in 2013. This is because a regular differences-in-differences analysis using nearby Western states as a control group would have limitations severely curbing its effectiveness. California is difficult to match with a single comparison state due to its size, economic structure, and population. The SCM addresses this issue by con-

structuring a weighted combination of states that approximates California’s pre-treatment characteristics. The synthetic control, as proposed by Abadie et al. (2003), is a weighted combination of 33 donor states, where weights are chosen to minimize the difference between California and our synthetic model during the pre-treatment period. These states are selected by starting with the contiguous 48 states, then eliminating any states that have similar climate policies to that of California. Including these states may contaminate the final analysis as they would likely have reduced carbon emissions. This means any state participating in the RGGI (Regional Greenhouse Gas Initiative) that aims to cap and reduce carbon emissions is left out of the donor pool. This includes the Northeast U.S. as denoted by the census, plus Virginia and excluding Pennsylvania. Washington and Oregon have similar cap-and-invest programs to that of California, meaning they are also left out of the pool. The state of Tennessee is excluded due to a highly unusual reliance on hydropower and nuclear energy, producing an energy mix not comparable to California’s. This may worsen pre-treatment fit however, as Tennessee closely matches California’s share of coal production.

Parameters included in the matrix of predictor states are population, GDP per capita, manufacturing share, energy mix (coal, gas, petroleum, renewable), and three emission lags. To capture the trajectory of emissions pre-treatment, the three lags are averaged across two periods (2001-2005 and 2006-2012). This practice reduces fluctuations that occur year to year. This predictor matrix is then used to estimate two synthetic controls, one using

MtCO₂e (Metric Tons of Carbon Dioxide Equivalent and one using real GDP growth. This allows us to evaluate if California’s cap-and-invest was successful in reducing carbon emissions while also not harming economic growth.

A diff-in-diff analysis is used to quantify the treatment effect by comparing post-treatment outcomes between California and the synthetic unit. This estimators for both analyses are obtained by using the following equations:

$$Emissions_{it} = \beta_0 + \beta_1 treated_i + \beta_2 post_t + \beta_3(treated_i \times post_t) + \varepsilon_{it} \quad (1)$$

$$GDP_{it} = \gamma_0 + \gamma_1 treated_i + \gamma_2 post_t + \gamma_3(treated_i \times post_t) + \varepsilon_{it} \quad (2)$$

$Emissions_{it}$ represents the treatment effect on emissions, while GDP_{it} represents the treatment effect on growth.

To quantify the significance of the results, I run placebo tests, as explained in Abadie et al. (2010), on both the synthetic controls and the DiD estimators. The placebo tests are performed by iterating over each state in the donor pool, assigning the treatment year to each donor state and then estimating a synthetic control. The MSPE ratio compares the post-treatment MSPE to the pre-treatment MSPE. It is then used to construct a pseudo p-value (the proportion of placebos whose MSPE ratio is higher than California’s). This helps us do inference on the results when lacking a sampling distribution that is random, such as in this case where there is just one unit (California). I then run the same placebo tests on the DiD estimators to obtain inferential statistics on how significant the results are.

One major limitation to this approach is that cap-and-trade is not the only climate policy implemented by California in the 21st century. These policies may result in reduced emissions well before the given treatment year of 2013. That being said, the SCM partially addresses this concern by matching the pre-treatment emissions trajectory, providing us an idea of how the state was trending before cap-and-invest.

Results

A synthetic model of emissions in California using our data returns the following table:

	Treated	Synthetic	Sample Mean
gdp_per_capita	56407.798	50513.291	47945.078
population	36059783.231	14604066.465	5675845.543
mfg_share	0.102	0.120	0.115
coal_share	0.010	0.123	0.335
gas_share	0.373	0.334	0.237
petrol_share	0.550	0.509	0.368
renewable_share	0.067	0.034	0.060
special.Emissions.2000.2005	384.933	389.968	134.302
special.Emissions.2006.2012	372.043	370.458	130.231

Figure 1: Synthetic Model of Emissions

While the synthetic model seems to be a good predictor, I should address a few limitations. For one, the elimination of certain states with a higher GDP per capita than California means that there is no state that can match California's actual value. The synthetic model comes about six thousand dollars short. More importantly however, is the clearly mismatched population

numbers. California is the largest state in the nation in terms of total population, making it impossible to find a state that can simulate its numbers. With that being said, the manufacturing share and energy mix efficiently minimized the distance between the predictor vector and California's vector, with the renewable share being a clear exception. The two emission lags are nearly identical, making us confident in our pre-treatment fit. We can further evaluate the pre-treatment fit of the synthetic model using the following:

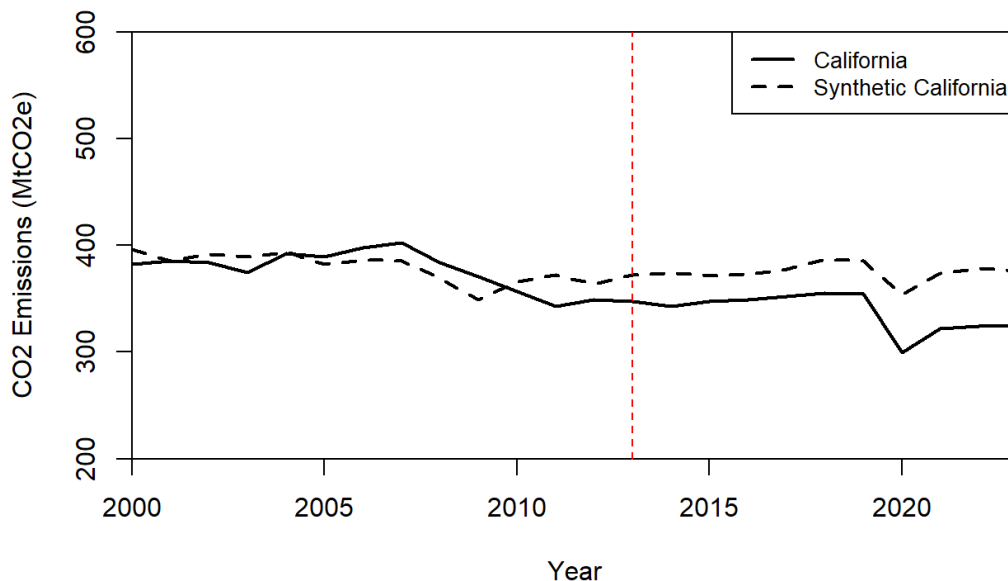


Figure 2: Synthetic vs. Treated Emissions

Here we get a visualization of the pre-treatment fit, with the red dotted line representing the treatment in 2013. This graph exposes a key limitation of this analysis: California's carbon emissions started to reduce a few years

before 2013. Throughout the late 2000s and early 2010s, the state passed multiple pieces of climate legislation targeting emissions reduction, many of which were implemented before cap-and-invest. We see this as the synthetic model follows California’s emissions trajectory fairly well up until 2009, where it suddenly diverges. This divergence raises a serious identification concern. The synthetic control method relies on a strong assumption that without treatment, the unit would have continued to follow the trend of the synthetic counterpart. The parallel trends assumption is likely violated due to other climate policies independent of the cap-and-invest program, policies that already targeted carbon emissions. This means that the estimated post-2013 reduction may reflect pre-existing trends rather than a specific causal effect relating to cap-and-invest.

After this, we move on to the placebo testing, which will assist us in inference. A plot of each placebo’s synthetic model returns the following:

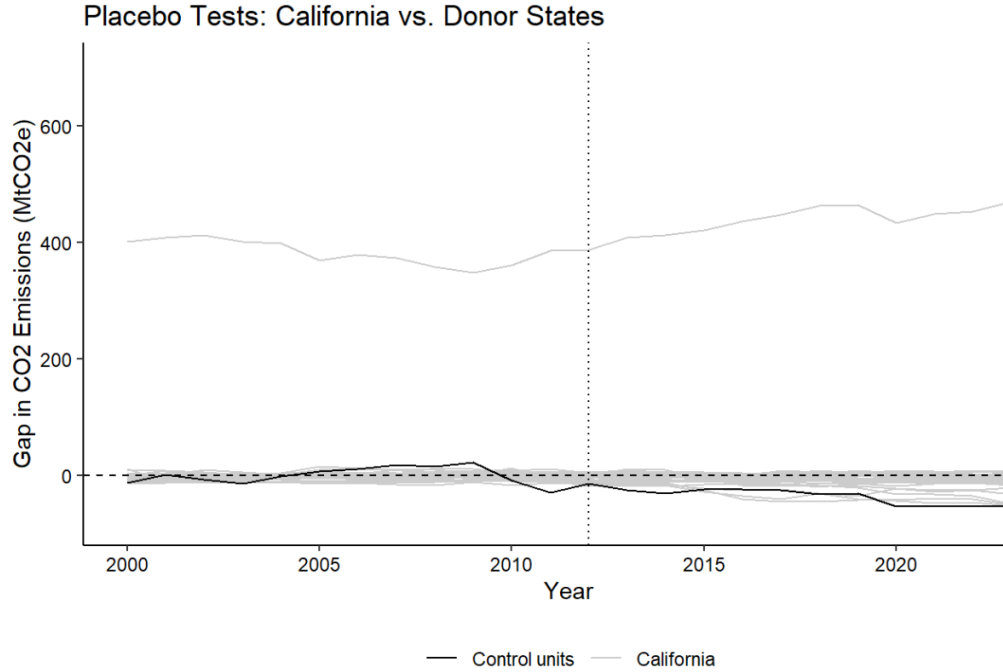


Figure 3: Placebo Tests

The labels are swapped in the graph, a quirk of the SCTools package in R. The black line is California’s model, while the gray lines are each control unit. The clear outlier with a poor pre-treatment fit here is identified as Texas, which will be dropped for the purpose of inference. The pseudo p-value obtained from this analysis is 0.875, meaning we cannot reject the null hypothesis. Using equation (1), we obtain a DiD coefficient of -35.41908, with a standard error of 8.910. This serves as the primary analysis, suggesting there was a reduction in carbon emissions. Once again using placebos, we obtain a one-sided p-value and two-sided p-value of 0.061 and 0.091 respec-

tively. This calls into question the statistical significance of the results. At a 5 percent level (as is convention), we cannot reject the null hypothesis and confidently say carbon emissions were reduced by cap-and-invest. However, it is still significant at a 10 percent level. This marginal significance suggests some generally weak evidence of a causal effect, but this should not be interpreted as robust, and instead be used cautiously. Overall, this indicates that the estimated treatment effect is not unusual relative to the placebo effects. The observed emissions reduction cannot be cleanly attributed to the policy. Moving on to the analysis of GDP growth, we construct our synthetic model using the same parameters, but replace emissions lags for averages of GDP growth in same years.

	Treated	Synthetic	Sample Mean
gdp_per_capita	56407.798	53454.315	47945.078
population	36059783.231	12134283.213	5675845.543
mfg_share	0.102	0.107	0.115
coal_share	0.010	0.195	0.335
gas_share	0.373	0.280	0.237
petrol_share	0.550	0.460	0.368
renewable_share	0.067	0.066	0.060
special.gdp_growth.2000.2005	3.600	3.588	2.996
special.gdp_growth.2006.2012	1.457	1.457	1.347

Figure 4: Synthetic Model of Growth

It is worth noting that this time, the synthetic GDP per capita is much closer to the treated. We also see that the pre-treatment is likely to be very strong, with most of the synthetic values closely matching their treated counterparts. The two most important values, the GDP growth averages

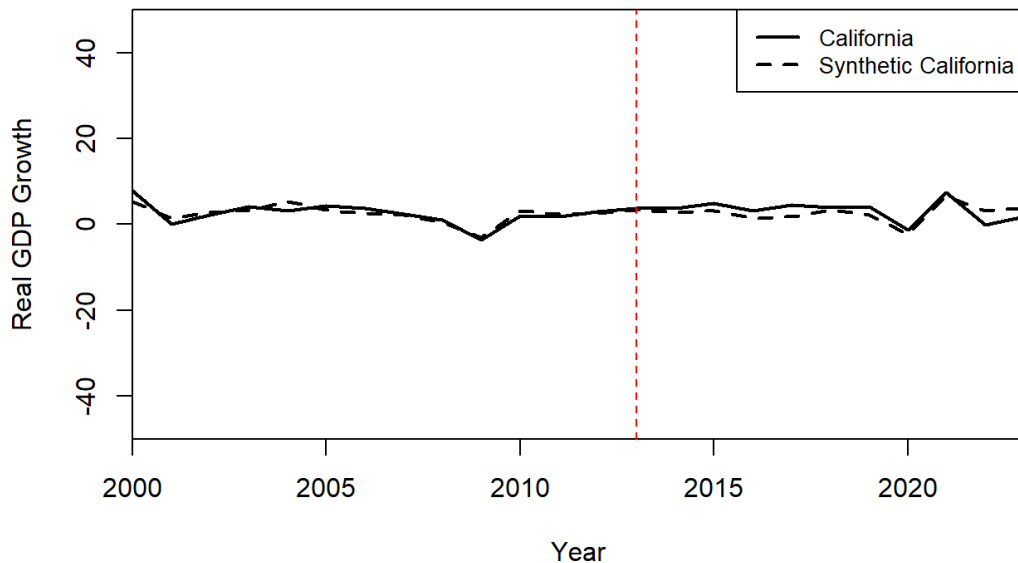


Figure 5: Synthetic vs. Treated Growth

between different time intervals, are virtually identical to the treated vector. Population is still notably an exception to this trend. It is subsequently unsurprising to see a strong pre-treatment fit when plotted. It is also to be expected that post-treatment, both trends match each other. This close pre-treatment fit suggests that the synthetic control is a good approximation of the counterfactual, despite it not ensuring a causal effect. Once again pivoting to placebo testing, we see a somewhat small gap in real GDP growth between California and its placebos. It is important to note again that the labels for California and the control units are swapped. This test returns a pseudo p-value of 0.455, meaning we again cannot reject the null hypothesis. Any changes in GDP growth that happened post-2013 are indistinguishable from noise, meaning there is no detectable impact of cap-and-invest on GDP

growth.

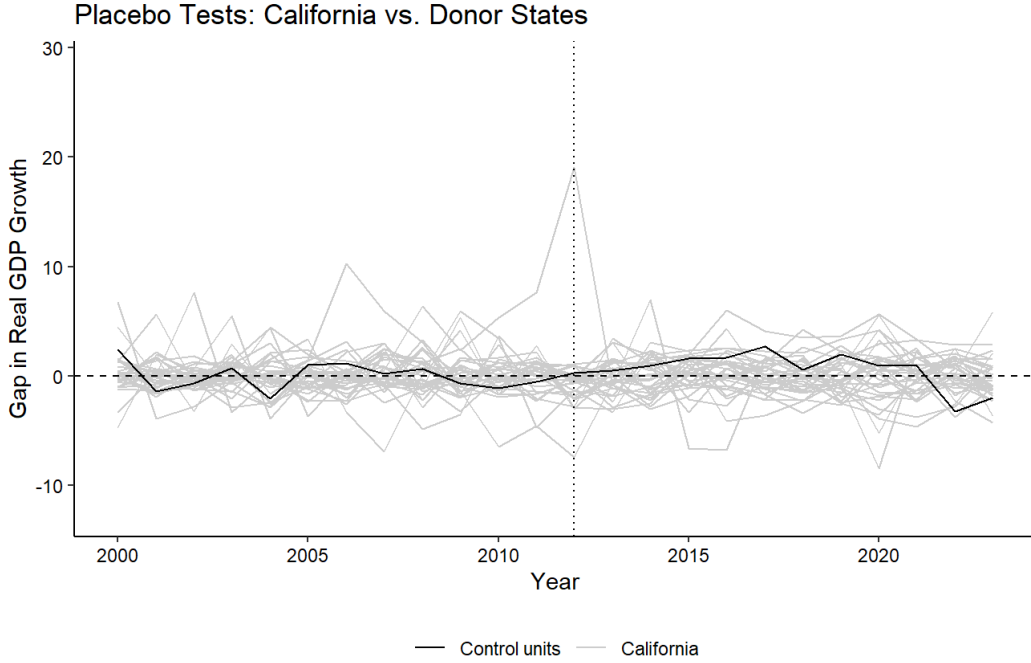


Figure 6: Placebo Tests

The diff-in-diff analysis returns results consistent with this conclusion, with our regression from equation (2) outputting 0.6044. The results are not particularly significant however, with our one-sided test returning 0.364 and two-sided test returning 0.515. This is statistically insignificant under any conventional significance level, meaning we cannot safely reject the null hypothesis that cap-and-invest had a causal effect on GDP growth. With everything in mind, the analysis provides limited evidence that cap-and-invest reduced emissions. Though the point estimates are certainly negative, our inference tests and the divergence that occurs pre-treatment make it

difficult to confirm a causal effect. Similarly, there is no evidence the program affected GDP growth.

Conclusion

This paper provided an analysis of the effect of California’s cap-and-invest program on carbon emissions and economic growth. Using a hybrid synthetic control framework and differences-in-differences approach, alongside placebo-based inference, we find limited evidence of a significant reduction in carbon emissions, and no evidence there was a meaningful impact on the state’s GDP growth.

While emissions did decline throughout the timeframe of the study, a significant divergence in pre-treatment fit and placebo tests indicate we cannot attribute these reductions safely to cap-and-invest alone. It is important to highlight with these findings, the difficulty that comes with evaluating a single climate policy that is part of a larger strategy to combat carbon emissions.

The results suggest that future policy should take into account that while market-based carbon approaches do not appear to harm economic growth, their effectiveness in deterring carbon emitting may be dependent on concurrent policies and sector-level analysis.

Overall, the paper is subject to several limitations, such as the difficulty in achieving a strong pre-treatment fit, which may obscure sector level dynamics

captured in previous literature. Future research may want to extend this analysis to the sectoral level, evaluate similar policies in other states, and be careful in the construction of counterfactuals. These findings underscore the importance of careful and reliable empirical methods when evaluating public policy.

References

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